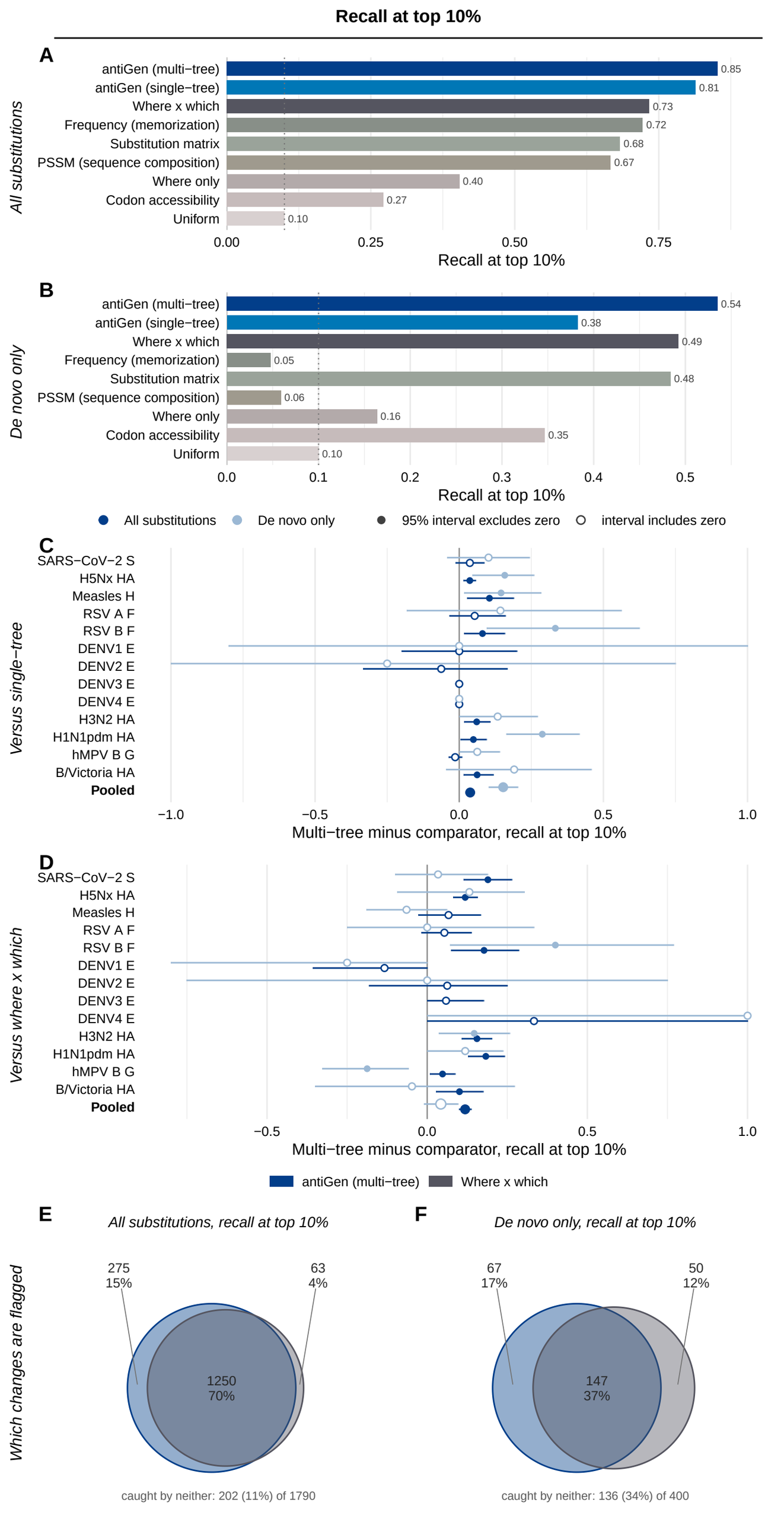
Recall at top 10%
A
All substitutions
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
PSSM (sequence composition)
Where only
Codon accessibility
Uniform
0.85
0.81
0.73
0.72
0.68
0.67
0.40
0.27
0.10
0.00
0.25
0.50
0.75
Recall at top 10%
B
De novo only
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
PSSM (sequence composition)
Where only
Codon accessibility
Uniform
0.54
0.38
0.49
0.05
0.48
0.06
0.16
0.35
0.10
0.0
0.1
0.2
0.3
0.4
0.5
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
−1.0
−0.5
0.0
0.5
1.0
Multi−tree minus comparator, recall at top 10%
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
−0.5
0.0
0.5
1.0
Multi−tree minus comparator, recall at top 10%
antiGen (multi−tree)
Where x which
E
All substitutions, recall at top 10%
F
De novo only, recall at top 10%
Which changes are flagged
275
15%
63
4%
1250
70%
67
17%
50
12%
147
37%
caught by neither: 202 (11%) of 1790
caught by neither: 136 (34%) of 400